TECHNICAL PAPER

PeopleSoft Agent

An AI assistant that works inside PeopleSoft HCM

Architecture · Features · End-to-end flows · Security · Operations

Product version	Widget 2026-11-02.5 · Agent service, November 2026
Tested on	PeopleSoft HCM 9.2 (HR92U052), PeopleTools 8.61, Oracle
AI model	Claude Opus 5.5 (claude-opus-5-5), Anthropic API
Author	Velu Angusamy
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Status	Working system; pre-production (see Chapter 7 for the open items)

PeopleSoft is a trademark of Oracle Corporation. Claude is a trademark of Anthropic. This paper describes an independent project and is not a product of either company.

About this paper

This paper describes the **PeopleSoft Agent**: a chat assistant that lives inside PeopleSoft HCM, does work on screen the way a person does, learns new tasks by watching them done once, investigates failed processes and access problems, watches the batch schedule overnight, loads spreadsheets through Component Interfaces and answers data questions from curated PS Queries. It explains how the system is built, what each feature does from end to end, how security is enforced, and how it is operated, tested and what remains to be done.

Who should read it

- **Architects and PeopleSoft administrators** who will install, run or extend the agent;
- **Security and audit reviewers** who need to know what the agent can see and do, as whom, and what is logged;
- **IT and HR leaders** who want to understand the capabilities, the cost and the risks before a rollout.

How to read the status labels

Label	Meaning
Proven	Run on the real PeopleSoft system (HR92U052, PeopleTools 8.61) with real Claude.
Partly proven	The main path was run on the real system; some parts only in tests.
Tested	Covered by automated tests against mock PeopleSoft pages and a scripted Claude only.

Conventions

Code, settings and PeopleSoft object names are in `monospace`. *USER, MONITOR, ADMIN* and *AUDITOR* are short for the PeopleSoft roles PSAGENT_USER, PSAGENT_MONITOR, PSAGENT_ADMIN and PSAGENT_AUDITOR. "The service" is the agent service; "the widget" is the browser part.

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1 Overview

1.1 What the agent is

PeopleSoft is powerful but demanding: work is spread over hundreds of components, batch failures need an expert to read logs and code, and security questions ("why can't she open this page?") mean walking a chain of PeopleTools tables by hand. The PeopleSoft Agent puts an assistant on every PeopleSoft page that can do this work with the user.

It is made of two parts. A JavaScript **widget** (`psagent.js`), loaded through PeopleSoft Branding, shows a chat panel and acts on the page in the user's own browser session. A Python **agent service** holds the conversation with Claude, the catalog of tasks, the rules and the audit log, and reads the PeopleSoft database read-only. The key idea is that **everything that changes PeopleSoft happens in the user's browser, as the user**, so PeopleSoft's own security applies, and nothing is saved without the user's click.

1.2 Design principles

These six rules apply to every feature. Where it matters, the service enforces them in code rather than relying on the model's behaviour.

#	Principle	How it is enforced
1	Nothing is saved, submitted or granted without the user's click.	Confirm cards and approval steps; the service refuses a save action that was not approved.
2	Passwords are never seen by the service or Claude.	Typed on a card in the browser and filled in there; never sent.
3	Troubleshooting only reads.	Read-only database access on three levels; investigations have no write tools.
4	Values are never invented.	Missing or unclear values are asked for or flagged; the service checks fields against the task.
5	Page, document, log and database text is data, never instructions.	Such text reaches Claude only as tool results, never as instructions; tools are narrow and checked.
6	Sensitive data stays masked, and every database query is logged.	Masking by column name; the audit log records every query with who and why.

1.3 Features at a glance

#	Feature	What the user gets	Who	Status
1	Chat widget	A panel on every page that knows the page and its error	All roles	Proven
2	Tasks	"Create a user profile for JSMITH": done on screen after a confirm card	USER	Proven
3	Recorder	Teach a new task by doing it once	ADMIN	Proven
4	Live mode	Work on pages that were never taught	USER	Proven
5	Live recovery	A recorded task finishes even if the page changed	USER	Tested
6	Documents	Attach a form; several people set up from one review table	USER	Tested

#	Feature	What the user gets	Who	Status
7	Process troubleshooter	"Why did process 557060 fail?" with root cause and fix	USER (own), MONITOR	Proven
8	Security troubleshooter	"Why can't ABSMITH open Create Job Opening?" and the smallest fix	USER (own), ADMIN	Proven
9	Absence reviewer	A recommendation and confidence score for a pending absence	USER (approver)	Proven
10	Runbooks	A checklist turned into a guided, logged run	ADMIN builds, USER runs	Proven
11	Watcher	Watchdogs overnight; every failure investigated; email with the analysis	MONITOR	Proven
12	Roles from PeopleSoft	Four agent roles on the normal user profile	All	Proven
13	Regression tests	Recorded tasks replayed after a patch or PUM image	ADMIN (test systems)	Partly proven
14	Feature bar	+ Features: guided flows by role	All	Tested
15	Admin dashboard	Usage, cost, audit log, failures, refused attempts	AUDITOR	Tested
16	Excel to CI	ExcelToCI without the macro workbook	USER	Partly proven
17	Ask the data	Questions answered from curated AI_ PS Queries	USER	Partly proven

1.4 What has been proven on the real system

- **Tasks and recorder:** Create User Profile, Sync Portal Security and running an Application Engine program; a user profile and a process run recorded and replayed.
- **Troubleshooting:** AEMINITEST 557060 (division by zero located to statement 3 in 21 seconds) and PORTAL_CSS 557046 (three runs at once with Max Concurrent 0).
- **Security:** ABSMITH (smallest role Hiring Manager, 62 components) and VELU (empty row-security table, fixed by a refresh).
- **Watcher:** AEMINITEST 557089 found and analysed in 43 seconds; 60 stuck requests found; a watch plan template read by Claude in 33 seconds.
- **Excel to CI:** all 457 CIs on HR92U052 through template, checks and request building; for 250 of them PeopleSoft found an existing record from exactly the keys the agent sends.
- **Ask the data:** the per-user query catalog, the run link, and a real 11,646-row answer (AI_USERS_WITH_ROLE) asked from the chat.

2 Architecture

2.1 Components

Figure 2-1 shows the parts and how they talk. Three channels matter: the **user's browser session** with PeopleSoft (blue, left), the **widget-to-service** API protected by a signed token (blue, middle), and the **service's own read-only access** to the database (red). The service talks to Claude over HTTPS and to an SMTP server for email.

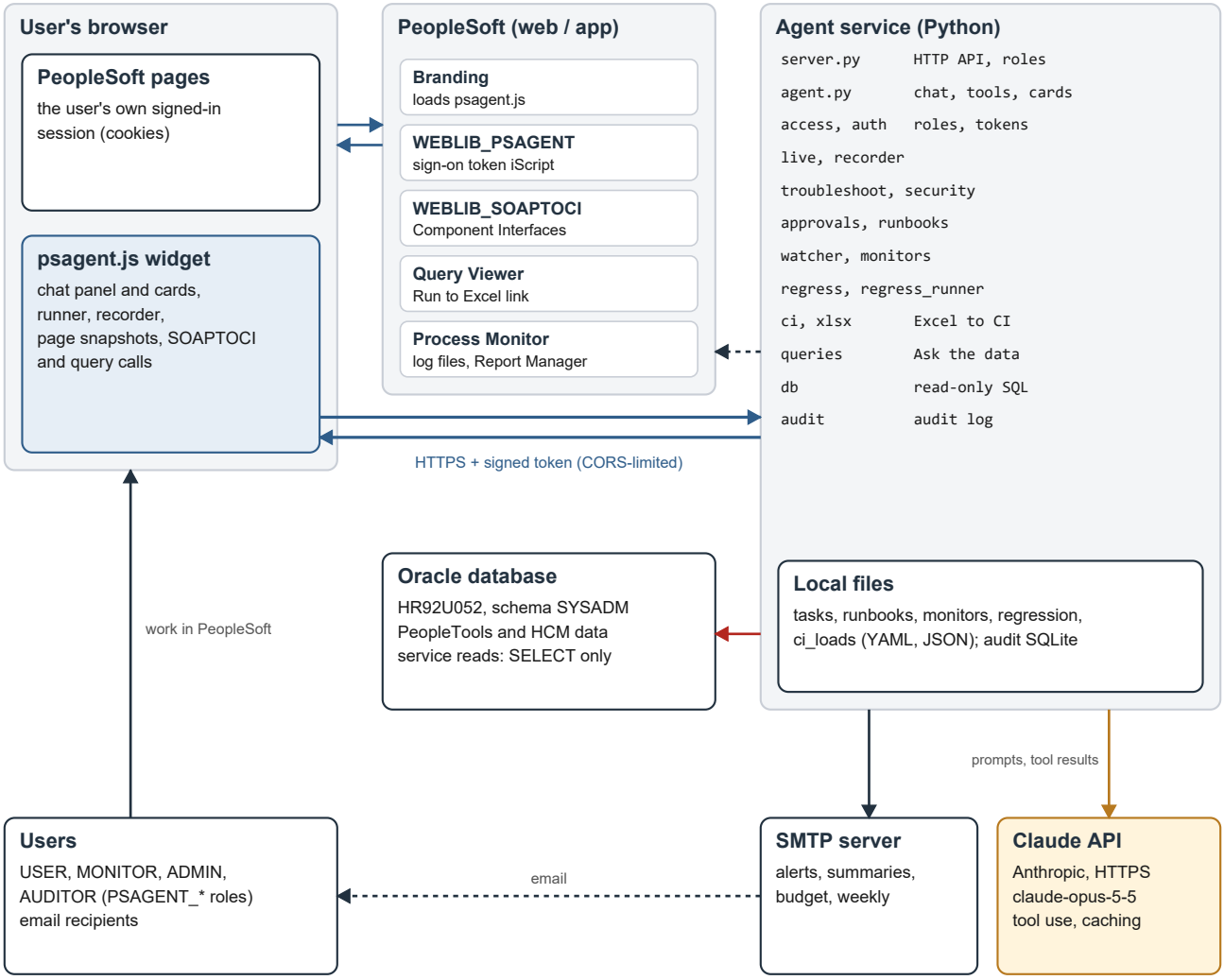


Figure 2-1 Components and channels. Blue: the user's own PeopleSoft session (left) and the token-protected API (middle). Red: the service's SELECT-only database reads. Dashed: log downloads as PSAGENT_SVC, and email.

2.2 The widget (psagent.js)

A single JavaScript file (about 4,100 lines, no external libraries) loaded on every PeopleSoft page through Branding. It runs in the page with the user's session, so every PeopleSoft call it makes is made as the user.

- **Panel:** chat, welcome tiles by role, the context bar ("You are on ..." and any error on the page), the activity feed, cards (confirm, diagnosis, review, results), the + Features bar and the watchdog indicator.
- **Runner:** performs a task's steps on classic pages: search pages (Add / Find), fields, grids, tabs, the Process Scheduler pop-up, and checks for Saved or a process instance.
- **Recorder:** records what a person does on a component (fields, clicks, grids, pop-ups, outcome) and sends it to the service to become a task.
- **Live mode:** sends a compact snapshot of the page (fields, buttons, tabs, messages) and performs the next action the service returns.
- **PeopleSoft calls in the user's session:** the sign-on iScript, View Log/Trace files, `WEBLIB_SOAPTOCI` for Component Interfaces, and Query Viewer's Run to Excel link.
- **Secrets stay local:** passwords typed on cards are filled into the page directly and never sent to the service.

2.3 The sign-on iScript (WEBLIB_PSAGENT)

`WEBLIB_PSAGENT.ISCRIPT1.FieldFormula.IScript_PSAgentConfig` returns the user's ID, name, agent roles, the service URL and a signed token. A user with no agent role gets no token and the widget stays hidden. The token and its checks are described in Chapter 3.

2.4 The agent service

A Python 3 application (Starlette on uvicorn) with one HTTP API used by the widget. It keeps conversations in memory, calls Claude with a fixed set of tools, checks every tool call, and writes an audit log. Table 2-1 lists its modules.

Module	Responsibility
server.py	HTTP API, token check, rights per request, refused attempts, start-up settings
access.py, auth.py	The four roles and what each may use; the token format and signature check
agent.py	The conversation with Claude: system prompt, tools, confirm cards, session state
llm.py	The Claude client (Anthropic SDK): effort, prompt caching, error handling, usage to the audit log
tasks.py, tasks/*.yaml	The task catalog: fields, validation, and the screen playbook for each task
recorder.py	Turns a recording into a task, with Claude's suggestions for the field names and questions
live.py	Live mode: the next action from a page snapshot; save actions need approval
troubleshoot.py	Process investigations: request, message log, logs, AE code, PeopleCode, overlapping runs
security.py	Access investigations: the security chain and the least-privilege fix
approvals.py	Absence request reviews: balance, overlaps, team coverage, history
runbooks.py	Drafting runbooks from checklists, checking them, running them step by step
monitors.py, watcher.py	Watchdogs and watch plans; the watcher loop, investigations and emails
regress.py, regress_runner.py	Regression suites and runs; the headless unattended runner

Module	Responsibility
ci.py, xlsx.py	Excel to CI: CI shapes, templates, row checks, SOAP requests, loads and results
queries.py	Ask the data: the user's AI_ query catalog, prompt checks, reading answers, summaries
db.py	Read-only database access: one SELECT, read-only transaction, rollback, limits, masking
audit.py	The audit log in SQLite: events, Claude usage and cost, dashboards, budget and weekly emails

Table 2-1 Agent service modules

2.5 Claude in the service

- **Model:** Claude Opus 5.5 (`claude-opus-5-5`) through the Anthropic API, with a configurable effort (low for chat, medium for investigations).
- **Tools, not free text:** Claude acts only through tools the service defines (26 in the chat, separate smaller sets for investigations and live mode). Each call is checked by the service before anything happens.
- **Strict tool schemas** make the API guarantee the shape of most tool inputs. The API allows at most 20 strict tools and limits their combined size; the service uses 17, and an automated test plus a live API test keep it within the limits.
- **Prompt caching:** the system prompt and tools are fixed per session, so repeated calls reuse the cache and cost less.
- **What Claude never receives:** passwords; masked sensitive columns; the full rows of a query answer (it gets totals, and at most 25 matching rows without sensitive columns when asked who has a value).

2.6 PeopleSoft integration points

Integration point	Used for	As whom
Branding (custom JavaScript)	Loading psagent.js on every page	Every user
WEBLIB_PSAGENT iScript	Sign-on token with the user's agent roles	The user
Classic pages	Tasks, live mode, recorder, security Apply	The user
View Log/Trace (Process Monitor)	Log files for the troubleshooter	The user (or PSAGENT_SVC overnight)
WEBLIB_SOAPTOCI iScript	Component Interfaces: GetCIShape, Create, Update	The user
Query Viewer Run to Excel link	Running the user's AI_ queries	The user
Fluid Approvals	Link from the absence review card	The user
Database (Oracle)	Read-only lookups: PeopleTools metadata, process requests, security tables, HCM data	The service's database account
PS_MAINTENANCE_LOG	Detecting a patch to start regression runs	The service's database account

2.7 What the service stores

Data	Where	Kept
Tasks, runbooks, watch plans, regression suites	YAML files: tasks/, runbooks/, monitors/, regression/	Until changed
Excel to CI loads (rows, results)	ci_loads/ (JSON)	Until removed
Audit log: events, queries, Claude usage	audit/psagent.db (SQLite)	90 days (PSAGENT_AUDIT_DAYS)
Conversations, query results, CI shapes in use	Memory only	Until the service restarts
Chat text, Claude's answers, documents, passwords	Not stored	Never

2.8 Deployment

In the tested setup the service runs on a Linux host (PeopleSoft's bundled Python works; `pysqlite3-binary` supplies SQLite where that Python lacks it) and is reached by the browsers over HTTPS. PeopleSoft itself is published through ngrok for testing. For production the recommended setup is a reverse proxy under the PeopleSoft host name, so the widget and the service share one origin.

```
uvicorn --factory psagent.server:app_from_env --host 0.0.0.0 --port 8000
```

Settings come from `agent-service/.env` (Section 6.2). The service refuses to start without the shared secret, and refuses to start if the development user is also set (Section 3.4).

3 Security

3.1 Trust model: who acts as whom

Actor	Identity	Can do
The widget	The signed-in PeopleSoft user, in their browser session	Everything the user can do in PeopleSoft, after the user's click for any save
The agent service	Its database account (read-only use) and its own secret	Read PeopleSoft data; never write; decide which agent features a user may use
Claude	None in PeopleSoft	Propose tool calls; every call is checked by the service; no direct access to anything
PSAGENT_SVC	A PeopleSoft service user	Download posted log files for the overnight watcher (Process Monitor and Report Manager only)
PSAGENT_TEST_USER (optional)	A PeopleSoft test user	Unattended regression runs on test systems

Why the browser does the work
Saving in PeopleSoft from the browser, in the user's session, means the user's page permissions, CI permissions, query security and row security all apply exactly as when they work by hand, and PeopleSoft's own audit fields show the real user. The service never holds a PeopleSoft password for the user and has no way to save on their behalf.

3.2 Sign-on and the token

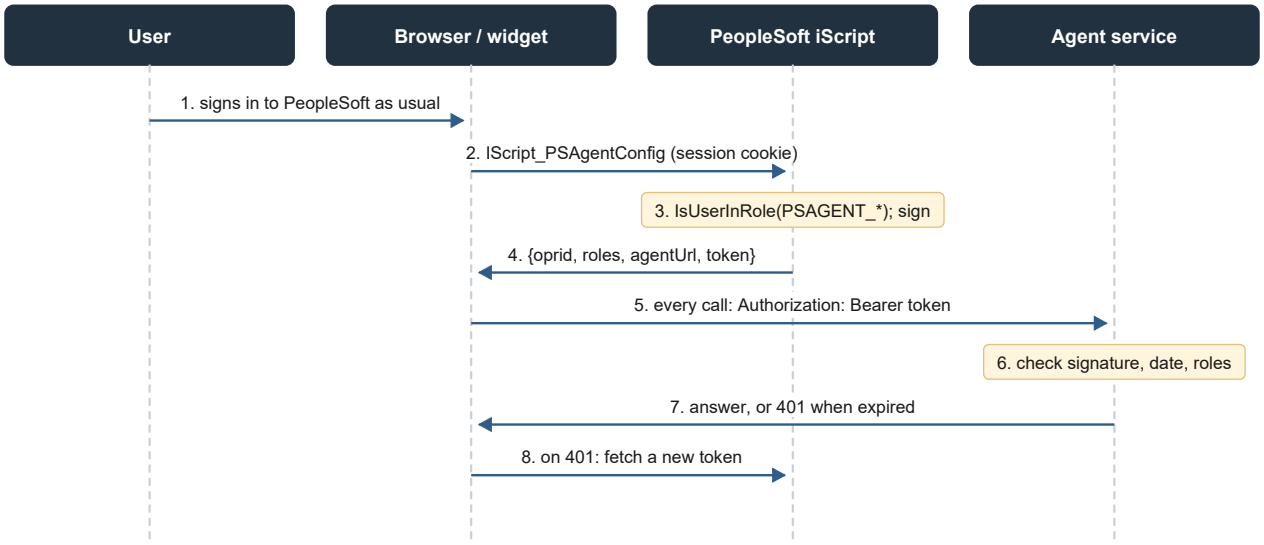


Figure 3-1 Sign-on: the token carries the user's identity and agent roles

The token is `OPRID.YYYYMMDD.ROLES.signature`, where the signature is SHA-256 of `secret|OPRID|YYYYMMDD|ROLES` computed by PeopleCode (`HashSHA256`). The secret exists only on the PeopleSoft application server and on the agent service. The service recomputes the signature, accepts the day of issue (plus or minus one day for time zones), and restricts the user ID to printable characters without the separator, which also blocks length-extension forgeries.

Known limitation: token lifetime
A token is valid for the calendar day. Until the browser fetches a new one (on page load or after a 401), it carries the roles the user had when it was issued, and a copied token works for the rest of the day. A shorter expiry (for example one hour, with the widget's silent refresh) is a small iScript change and is recommended before production.

3.3 Roles

Access to the agent is managed on the normal PeopleSoft user profile with four roles. USER, MONITOR and ADMIN build on each other; AUDITOR stands apart so that administrators do not audit themselves (a person may hold both).

Feature	USER	MONITOR	ADMIN	AUDITOR
Chat, explain a page or an error	●	●	●	●
Run recorded tasks, live mode, documents	●	●	●	
Run published runbooks	●	●	●	
Ask the data (AI_ queries)	●	●	●	
Excel to CI	●	●	●	
Absence reviewer, own approvals	●	●	●	
Troubleshoot own processes	●	●	●	
Troubleshoot anyone's processes, with Claude's own read-only queries		●	●	
Watcher: watchdogs, alerts, manage		●	●	
Check my own access	●	●	●	
Check anyone's access, and Apply a fix			●	
Record tasks, publish runbooks			●	
Regression tests (test systems)			●	
Everyone's absence requests			●	
Admin dashboard and audit log				●

3.4 How access is enforced

- **On every request, by the service.** Each request's rights are computed from its signed token (`access.py`), not remembered from the first message. A refused feature answers 403; a refused tool tells Claude it is not allowed, and Claude tells the user which role is needed. The widget only hides what a user cannot use; hiding is not the control.
- **Ownership checks inside features.** A USER may troubleshoot only their own processes and check only their own access; other users' runs of the same process are shown without the user and run control. Investigations for a USER use fixed lookups only; Claude's own read-only queries are for MONITOR and ADMIN.
- **No role, no agent.** A token without any agent role is refused, and the iScript gives such a user no token.
- **Development mode cannot reach production.** With `PSAGENT_DEV_OPRID` set, requests without a token run as that user. The service refuses to start if it is set together with the shared secret, and the development lists of teachers, monitors and auditors are ignored outside development.

3.5 PeopleSoft's own security still applies

The agent role decides which features a user sees. Inside each feature PeopleSoft decides what the user can do: tasks and live mode use the user's page permissions; Excel to CI uses their Component Interface permissions; Ask the data offers only queries whose every record is in the user's query trees, and PeopleSoft applies their row security when it runs the query. The agent cannot give a user more access than PeopleSoft does.

3.6 Protecting data

- **Passwords** are typed on a card and filled into the page by the browser; they are not sent to the service, Claude or the audit log.
- **Sensitive columns** (national IDs, birth dates, pay rates and amounts, bank details, passwords) are masked in every database result by column name before Claude sees it.
- **Ask the data** sends Claude the row count and per-column counts and totals; to name who has a value it sends at most 25 matching rows, never sensitive columns, and refuses to filter or total on them.
- **Chat text and answers are not stored**; the audit log records that a message was sent, not what it said. Documents are read and discarded.
- **Prompt injection:** page text, documents, logs and data reach Claude as tool results, never as instructions; tools are narrow; a save needs the user's click enforced by the service; the database is read-only.

3.7 Read-only database access

Even with a privileged database account, the service cannot write. Every statement is checked to be a single SELECT or WITH; every session runs `SET TRANSACTION READ ONLY`; every query is rolled back. Results are capped at 200 rows and 20 seconds, and every query is logged with the user and the reason. A dedicated SELECT-only account (`PSAGENT_RO`) is planned to replace SYSADM.

3.8 Audit and refused attempts

Every Claude call (tokens, estimated cost), database query (SQL, row count), task run (confirmed values), save, publish, watchdog change, investigation, alert and regression run is recorded with the user and the feature. Refused attempts are recorded as `access.denied`: a feature or tool used without the role, another user's process, access, query or absence request, a token without an agent role, and a forged or malformed token. The dashboard counts them and the audit log filters them. Expired tokens are normal and not counted.

3.9 Threats and mitigations

Threat	Mitigation	Residual risk
Someone calls the service as another user	Signed token; secret only on the app server and service; 401 and an audit entry	A leaked token works until the end of the day
A user reaches a feature their role does not allow	Rights computed per request from the token; tools refused by the service	Role changes apply when a new token is fetched
Claude is tricked by page, log or document text	Text is data; narrow, checked tools; saves need the user's click; read-only database	Claude may give a wrong answer; nothing is changed without the user
The service writes to the database	Single SELECT, read-only transaction, rollback	SYSADM account until PSAGENT_RO exists

Threat	Mitigation	Residual risk
Sensitive data reaches Claude	Masking by column name; totals instead of rows; documents discarded	A sensitive column with an unusual name is not recognised
A user runs a query outside their query security	Catalog from query trees; links built only for catalog queries; PeopleSoft row security	Query Viewer's link itself does not check query trees (to be retested)
Development mode left on in production	Start-up refused when the dev user and the secret are both set	None known
Another website calls the service	CORS limited to the PeopleSoft origin; token required	None known
Runaway Claude cost	Cost per call in the audit log; monthly budget emails; cap on overnight investigations; prompt caching	Budget emails need SMTP
Service host compromised	Secrets only in .env; read-only use of the database	Keys in a file, not a vault; SYSADM account

4 Features

Each section gives the purpose, how the feature works, who can use it, its safeguards and what has been proven. Chapter 5 shows the main features as end-to-end sequence diagrams.

4.1 Chat widget

Who	Every user with an agent role	Status	Proven
Claude used	For chat answers		

A wide panel docked on the right of every PeopleSoft page, opened with Ctrl+/ or the launcher.

How it works

- **Welcome screen:** tiles for what the user can do (by role and tasks), recent questions, overnight alerts for monitors.
- **Context bar:** the page the user is on and any error shown on it, with *Explain this page* and *Help with this error*.
- **Activity feed and avatar:** every step the agent takes, with timings; ready, thinking, working, waiting, stopped.
- **History:** kept for the browser session; wide mode for large cards.

Safeguards

- Hidden for users without an agent role (no token).

Proven

Loaded through Branding on HR92U052; sign-on through the WEBLIB_PSAGENT token.

4.2 Tasks: the agent does the work

Who	USER	Status	Proven
Claude used	To collect the values; no Claude during the run		

"Create a user profile for JSMITH, roles EMPLOYEE and Manager." A task is a recorded PeopleSoft procedure with its fields and screen steps (tasks/*.yaml).

How it works

- Claude picks the task and collects the values in the chat, asking only for what is missing.
- The service validates every value against the task definition.
- A **confirm card** lists everything; passwords are typed on the card.
- After **Confirm** the widget performs the steps on screen: search or add, fields, grids, tabs, the Process Scheduler pop-up.
- It stops on any PeopleSoft error and explains it; otherwise it reports Saved or the process instance.

Safeguards

- Nothing runs before Confirm; changing a value withdraws the card.
- IDs, roles and permission lists are never invented.
- The run is in the user's session with their permissions.

Proven

Create User Profile, Sync Portal Security and Run an Application Engine program on HR92U052; about $0.02 per run.

4.3 Recorder: teach a task by doing it once

Who	ADMIN	Status	Proven
Claude used	Once, to suggest field names and questions		

An administrator clicks **Record**, does the task once by hand, and the agent turns the recording into a reusable task.

How it works

- Understands search pages (Add / Find / already exists), transaction pages, grids, pop-ups such as the Process Scheduler request, and the outcome.
- Leaves out edits the user undid and clicks that did nothing.
- Claude suggests field names, which fields to ask for and which are fixed or optional; the teacher reviews and saves.
- The task is available at once, to every USER allowed by PeopleSoft to use that page.

Safeguards

- Only ADMIN may record or overwrite tasks; the recorded task never stores passwords.

Proven

A user profile and a process run recorded and replayed on the real system.

4.4 Live mode: pages it was never taught

Who	USER	Status	Proven
Claude used	One call per page step		

When no task fits, or the user says *"do it live"*, Claude operates PeopleSoft one page at a time.

How it works

- The widget sends a snapshot of the page: fields, buttons, tabs, messages and pop-ups.
- Claude chooses the next action; the widget performs it and sends the next snapshot.
- Missing values are asked for; passwords go on a card.

Safeguards

- Any action that saves, submits, deletes or posts needs the user's approval click, enforced by the service.
- Classic pages only today.

Proven

On real classic pages; about $0.015 to $0.02 per step.

4.5 Live recovery

Who	USER	Status	Tested
Claude used	When a recorded step fails		

If a page has changed since a task was recorded (a moved field, a new tab), the recorded run hands over to live mode at the failed step.

How it works

- Live mode finishes the task; the save is approved again by the user.
- The agent then offers to update the task from what it learned.

Proven

Browser test with a moved field; not yet seen on a real page change.

4.6 Documents: a form fills PeopleSoft

Who	USER	Status	Tested
Claude used	To read the document		

Attach a PDF, an image or a spreadsheet, such as an access request form listing several people.

How it works

- Claude reads the document and maps it to the right task.
- Handwritten, ambiguous or described values are flagged for checking.
- Several people appear in one review table; the user fixes, fills in or unticks rows and the service checks every edit.
- Rows run one by one with a result each.

Safeguards

- Documents are read and discarded, never stored.

Proven

Browser test with a two-person access form.

4.7 Process troubleshooter

Who	USER (own processes); MONITOR and ADMIN (anyone's)	Status	Proven
Claude used	An investigation of 20 to 60 seconds		

"Why did process 557060 fail?" or *"why did my last process fail?"* The agent investigates like a PeopleSoft administrator, read-only.

How it works

- **Log files** (stdout, .log, .trc) read in the user's browser session, the same files as View Log/Trace; only the end of very large files.
- **Request details:** parameters (passwords hidden), the message log with the text filled in, earlier runs.
- **Code:** Application Engine sections, steps and SQL; PeopleCode with statement numbers matching "Statement:n"; a search of all PeopleCode for functions defined elsewhere, including delivered code.
- **Runs at the same time:** overlapping runs, Max Concurrent, restart settings, mutual exclusion.
- **Diagnosis card:** category and confidence, exact place, root cause, evidence, fix steps and corrected code, how to rerun, every query it ran, Copy report.

Safeguards

- A USER may investigate only their own processes; Claude's own queries are for MONITOR and ADMIN.
- Nothing is rerun or changed.

Proven

AEMINITEST 557060 (division by zero at statement 3, 21 seconds) and PORTAL_CSS 557046 (three runs at once, Max Concurrent 0).

4.8 Security access troubleshooter

Who	USER (own access); ADMIN (anyone's, and Apply)	Status	Proven
Claude used	An investigation of about 25 seconds		

"Why can't ABSMITH access Create Job Opening?" The agent walks the security chain from the PeopleTools tables.

How it works

- The page to its component (MENU.COMPONENT.MARKET); the user profile (exists, locked), roles and permission lists.
- The component's pages and actions against what the permission lists authorise; the menu link permission.
- Row security on the search record, including the menu item's override; HCM row security (SJT tables).
- **Least-privilege fix:** every role that would grant the access, with how many other pages it would also give and how many users it affects.
- **Apply** (ADMIN): a live run makes exactly that change and stops for approval before Save.

Safeguards

- Nothing is ever granted automatically.
- A USER can check only their own access.

Proven

ABSMITH lacks HCCPRS1040; the smallest role, Hiring Manager, adds 62 components against 81 to 178 for the others. VELU has an empty SJT_OPR_CLS: the fix is a refresh, not a new role.

4.9 Absence approval reviewer

Who	USER (their approvals); ADMIN (everyone's)	Status	Proven
Claude used	A review of 30 to 45 seconds		

The approver lists the absence requests waiting for them and clicks **Review with AI**.

How it works

- Checks from the database, read-only: the request, the entitlement balance, overlaps, team coverage under the same manager, timing, employment status and history.
- The card shows PeopleSoft's approval status, a recommendation (approve, send back, deny, needs your review), a confidence score 0 to 100 with reasons, each check and a suggested comment.

Safeguards

- The agent never approves; the approver decides in PeopleSoft (Open Approvals).
- An approval despite a failed check is capped at 50.

Proven

Request 1589: all checks fine but the date 8 years ago, so "needs your review" at 65/100, in 42 seconds.

4.10 Runbooks from a checklist

Who	ADMIN builds and publishes; USER runs	Status	Proven
Claude used	To draft; not during a run		

An administrator attaches the checklist they already have (Excel, CSV or PDF) and asks for a runbook.

How it works

- Claude turns each row into a step: a check (read-only query), a task or process from the catalog, an approval or a manual step; values that change each run become inputs.
- The service checks tasks and fields exist, stores no passwords, and test-runs every check query right away.
- **Publish** saves runbooks/<id>.yaml; a *Run:* tile appears.
- A run asks for the inputs, then: checks run on the service, tasks show their confirm card, processes are followed in Process Monitor, approvals and manual steps wait for a click.

Safeguards

- The run stops at the first problem; every run is logged with who, inputs and each step's result.

Proven

A 5-step checklist drafted by real Claude in 30 seconds; a process followed to Success (PORTAL_CSS 557093).

4.11 Watcher: watchdogs and alert emails

Who	MONITOR	Status	Proven
Claude used	Only to investigate a new failure		

Watchdogs run on the service day and night, with nobody signed in.

How it works

- **Four kinds:** failure (Error or No Success), stuck (still queued N minutes after the run time), must succeed by (a time of day; catches a job that never ran), and data check (a read-only query returning wrong rows).
- Set up from an Excel template (watch plans) or one sentence in the chat; a preview shows what each would have caught in the last 30 days.
- Every minute: one read-only query per due watchdog, no Claude.
- Every new failure is investigated by the troubleshooter, with log files downloaded as PSAGENT_SVC; repeats within 12 hours are counted, not re-investigated; investigations are capped per day.
- The alert email gives what happened, the analysis and the action needed, with links; a daily summary goes to the team.

Safeguards

- The watcher only reads; nothing is rerun or changed.

Proven

AEMINITEST 557089 found and analysed in 43 seconds; 60 stuck requests found; a 4-row plan read by Claude in 33 seconds.

4.12 Roles from PeopleSoft

Who	Everyone	Status	Proven
Claude used	No		

The four agent roles are ordinary PeopleSoft roles on the user profile, signed into the token (Chapter 3).

Proven

USER, MONITOR and ADMIN on HR92U052; the AUDITOR role and the updated iScript are to be deployed.

4.13 Regression tests

Who	ADMIN, test systems only	Status	Partly proven
Claude used	Only to finish a case live when a page changed		

Every recorded task is an exact replay, so on a test system it becomes a regression test after a patch or a PUM image.

How it works

- A smoke test of every task with no setup; or a test plan from an Excel template with expected results (Saved, Process submitted, or an expected Error for negative tests).
- Results: Passed; Passed with changes (the page changed and the case was finished live, with the change listed); Failed.
- Compared with the last run; reports as Excel; run in the browser or unattended in a headless browser on the service, daily or when PS_MAINTENANCE_LOG changes.

Safeguards

- Off unless PSAGENT_TEST_ENV=on, because the cases really save data.

Proven

Browser tests of all three results; the headless runner signed in to PeopleTools 8.61.

4.14 Feature bar

Who	Everyone, by role	Status	Tested
Claude used	Depends on the flow		

+ Features under the message box lists the guided flows the user's roles allow: watch jobs, manage watchdogs, regression tests, runbooks, troubleshoot, check access, absence approvals, record a task, ask the data, load with a CI, audit.

4.15 Admin dashboard and audit log

Who	AUDITOR	Status	Tested
Claude used	Only when an auditor asks about it in the chat		

Opens on the left of the screen with the chat on the right.

How it works

- **Overview:** people, conversations, task runs, Claude cost against the period before, failures, refused attempts, alerts, the month's spend against the budget, cost per day, by feature and top users.
- **Usage and cost** by user or feature; **audit log** with filters (user, feature, kind, outcome, dates, words) and the SQL or confirmed values per row; **failures**; **watcher and tests**. Exports to Excel.
- Budget emails at 80% and 100% of PSAGENT_BUDGET; a weekly usage email.

Safeguards

- Chat text, answers and passwords are not kept; events are kept 90 days.
- Cost is an estimate from tokens at list prices.

Proven

Browser test with 30 days of seeded history; real data starts after the service restart.

4.16 Excel to CI

Who	USER (PeopleSoft's CI permissions decide which CIs)	Status	Partly proven
Claude used	No (only "Explain the errors")		

ExcelToCI's job without the macro workbook: pick a Component Interface, download a template, upload it filled in, and load.

How it works

- The CI's shape is read with GetCIShape through the delivered WEBLIB_SOAPTOCI iScript (as ExcelToCI's "New Template"), plus PeopleTools metadata: prompt tables, translate values, Yes/No, required fields, create and get keys.
- The template: the CI's sheet (Action, keys, chosen properties), one sheet per collection linked by the parent keys, a Help sheet, drop-downs.
- Every row is checked before anything is saved: keys, required fields, lengths, numbers, dates and times, translate and prompt values (read-only queries), Create of an existing key, Update of a missing one, duplicates, orphans.
- The browser sends each row to WEBLIB_SOAPTOCI in the user's session; results per row; Pause, Stop, Resume; the rows that failed download as a template to fix.

Safeguards

- Saved as the user with their CI permissions; the CI's own PeopleCode applies.
- Passwords are never in the file.
- Every row is in the audit log.

Proven

All 457 CIs on HR92U052 through template, checks and requests; the update keys found existing records for 250 CIs (read-only); formats confirmed from the delivered SOAPTOCI code. No real Create or Update has been run yet.

4.17 Ask the data

Who	USER (their AI_ queries and row security)	Status	Partly proven
Claude used	Two to three chat turns per question		

"How many new hires this year by department?" answered only from PS Queries named AI_..., which an administrator writes in Query Manager. The agent never writes SQL for data questions.

How it works

- The catalog: public AI_ queries and the user's own private ones whose every record is in the user's query trees.
- Claude picks the query and fills its prompts from the question; the service checks every value (dates as YYYY-MM-DD, translate and prompt values).
- The browser runs the query with Query Viewer's Run to Excel link in the user's session; PeopleSoft applies row security.
- The service reads the answer; the user sees a card with the rows (first 200), a chart, Download Excel and Open in Query Viewer.
- Claude gets totals; for "who has X" it filters the whole result and gets at most 25 matching rows without sensitive columns.

Safeguards

- Links are built only for queries in the user's catalog.
- Sensitive columns are never shown to Claude, totalled or filtered on.
- Results stay in memory for the conversation.

Proven

The catalog per user, the link with prompts, the date format and real answers (324 and 11,646 rows).

5 End-to-end flows

The diagrams use five colours of arrow: blue between the user, the widget, PeopleSoft and the service; brown to and from Claude; red for the service's read-only database queries. Yellow boxes are work done inside one participant.

5.1 A recorded task

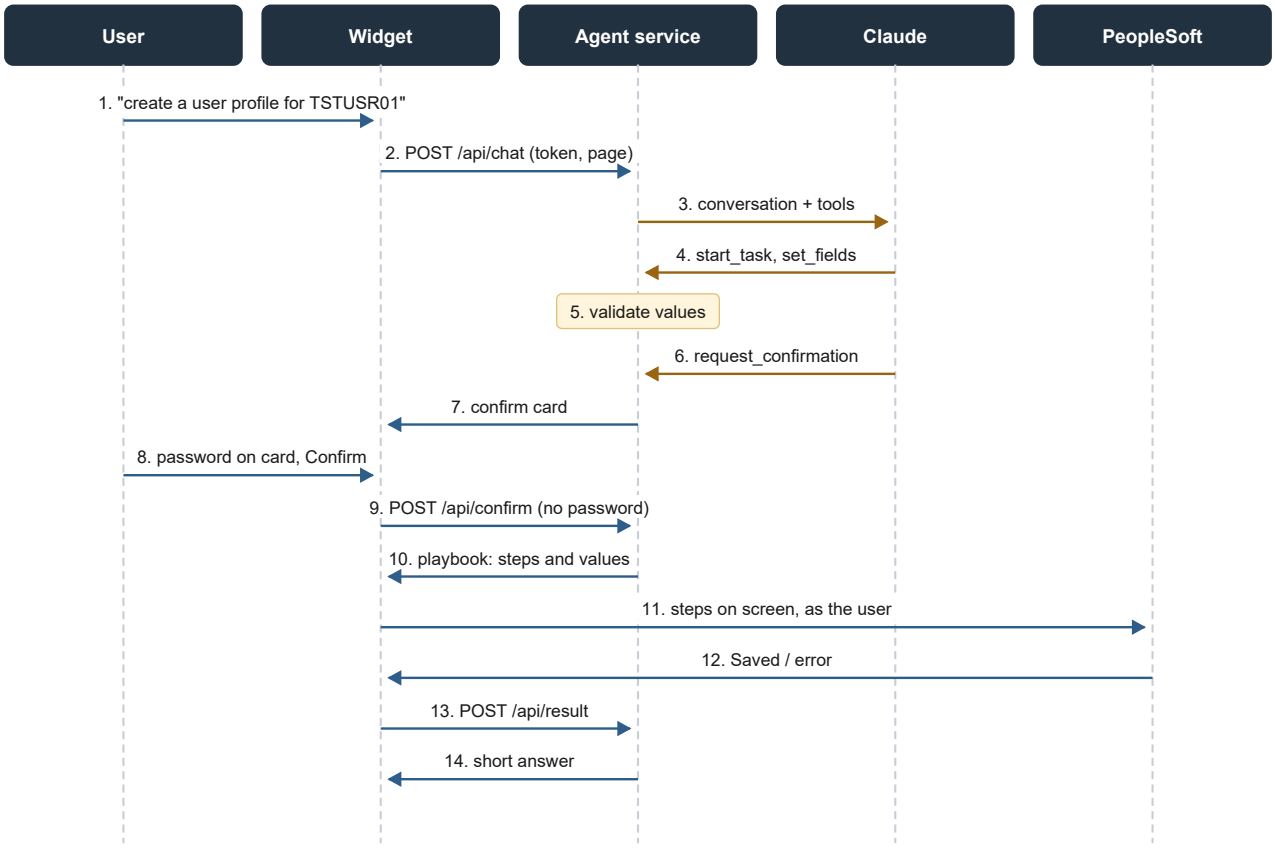


Figure 5-1 A recorded task from request to Saved

5.2 Live mode

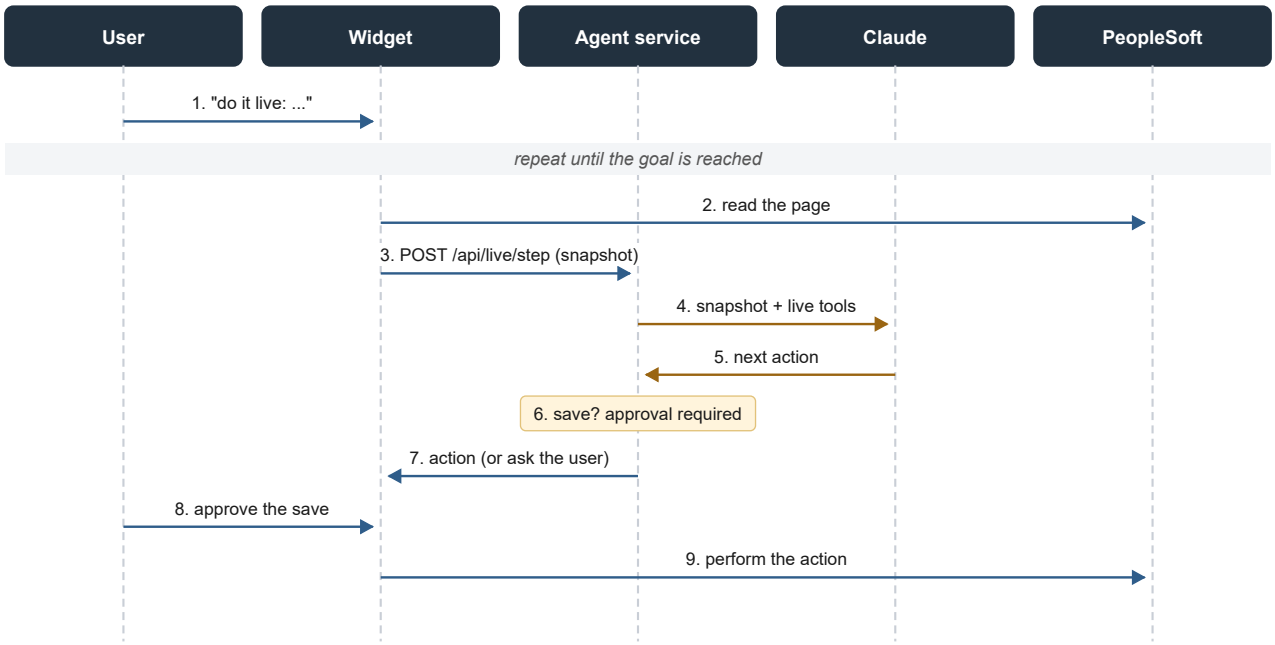


Figure 5-2 Live mode: one Claude call per page step; saves need approval

5.3 Process troubleshooting

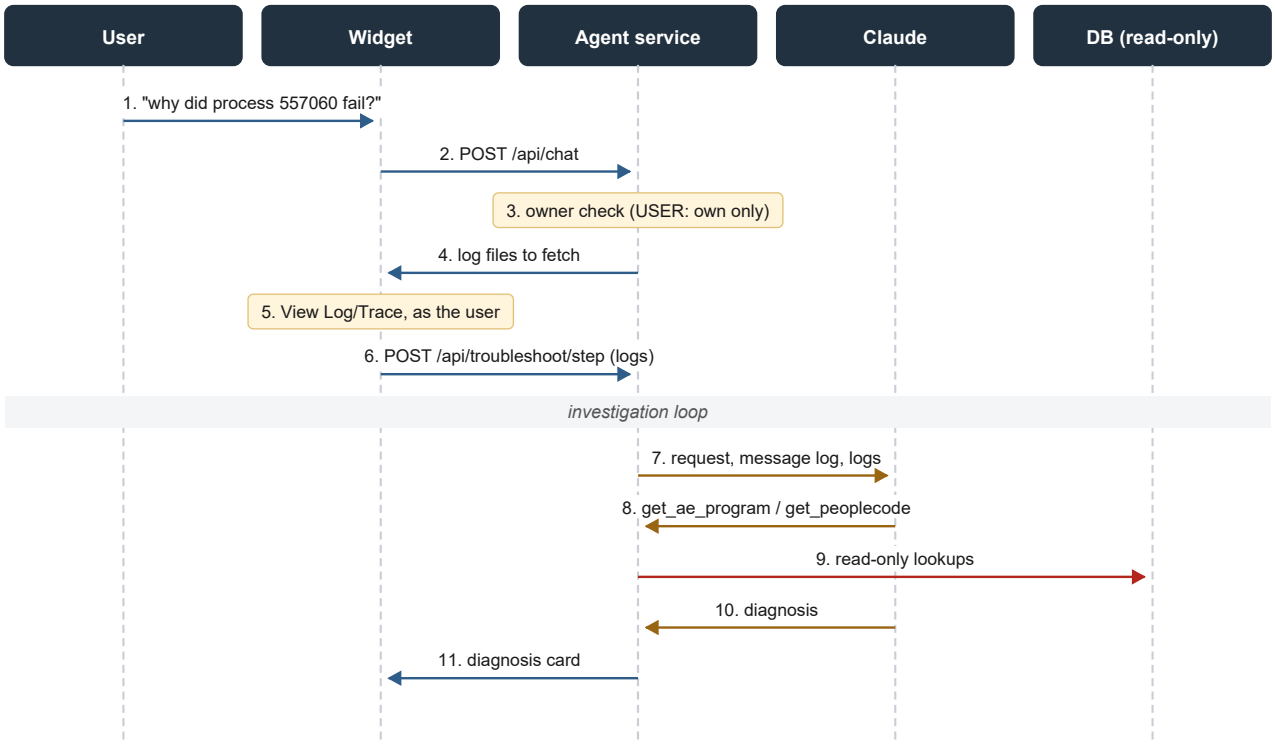


Figure 5-3 Investigating a failed process

5.4 The watcher overnight

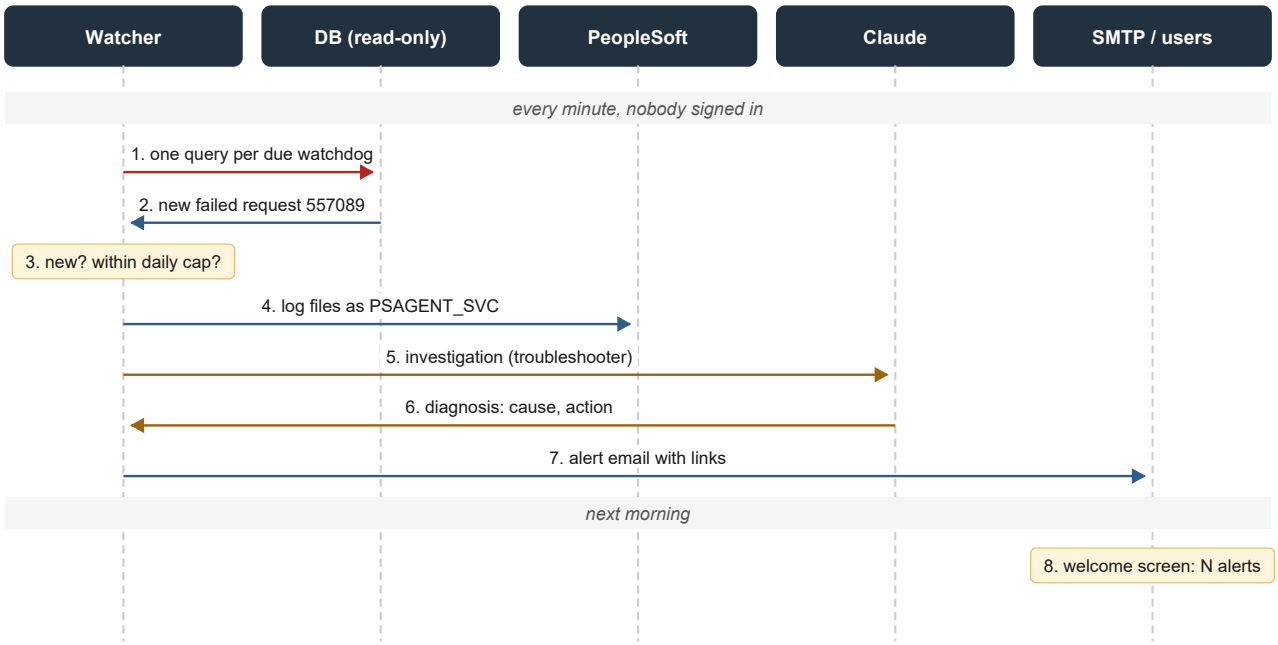


Figure 5-4 A failure found, investigated and emailed overnight

5.5 Excel to CI

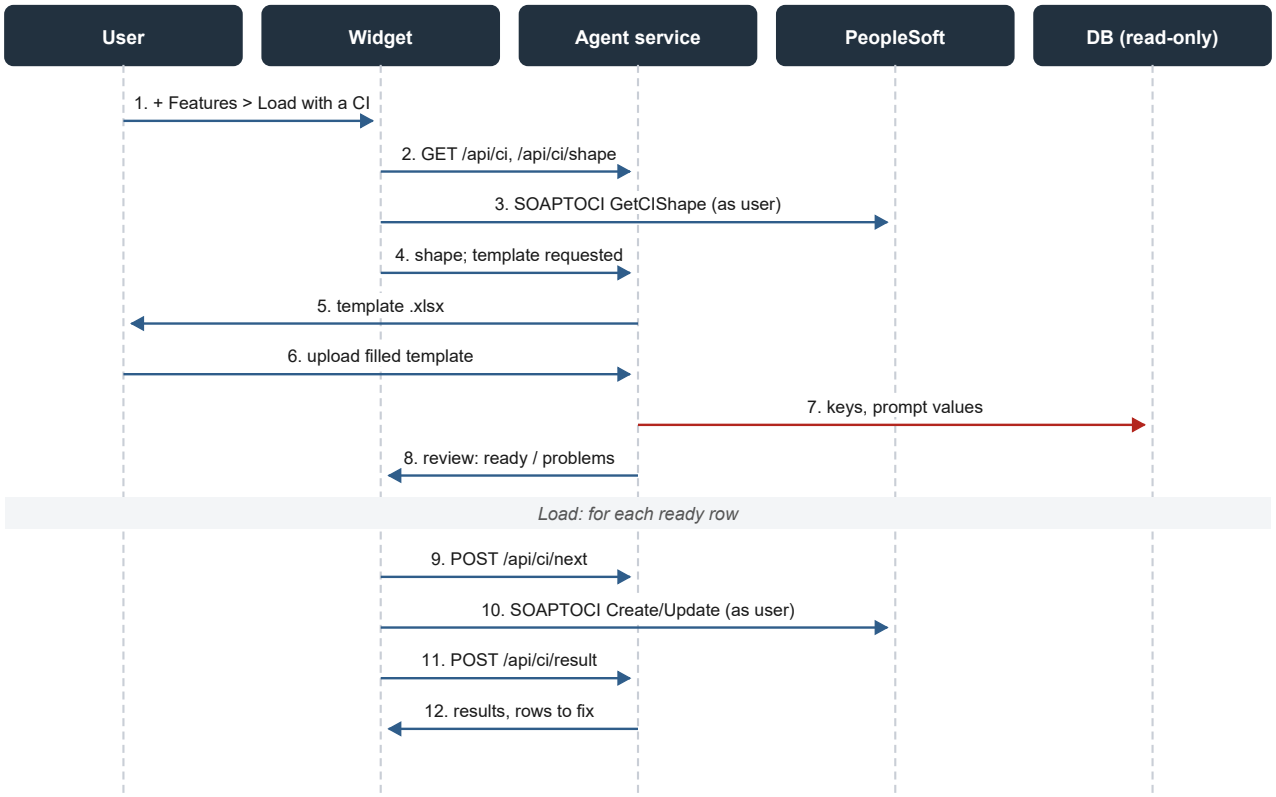


Figure 5-5 Excel to CI: no Claude in the checks or the load

5.6 Ask the data

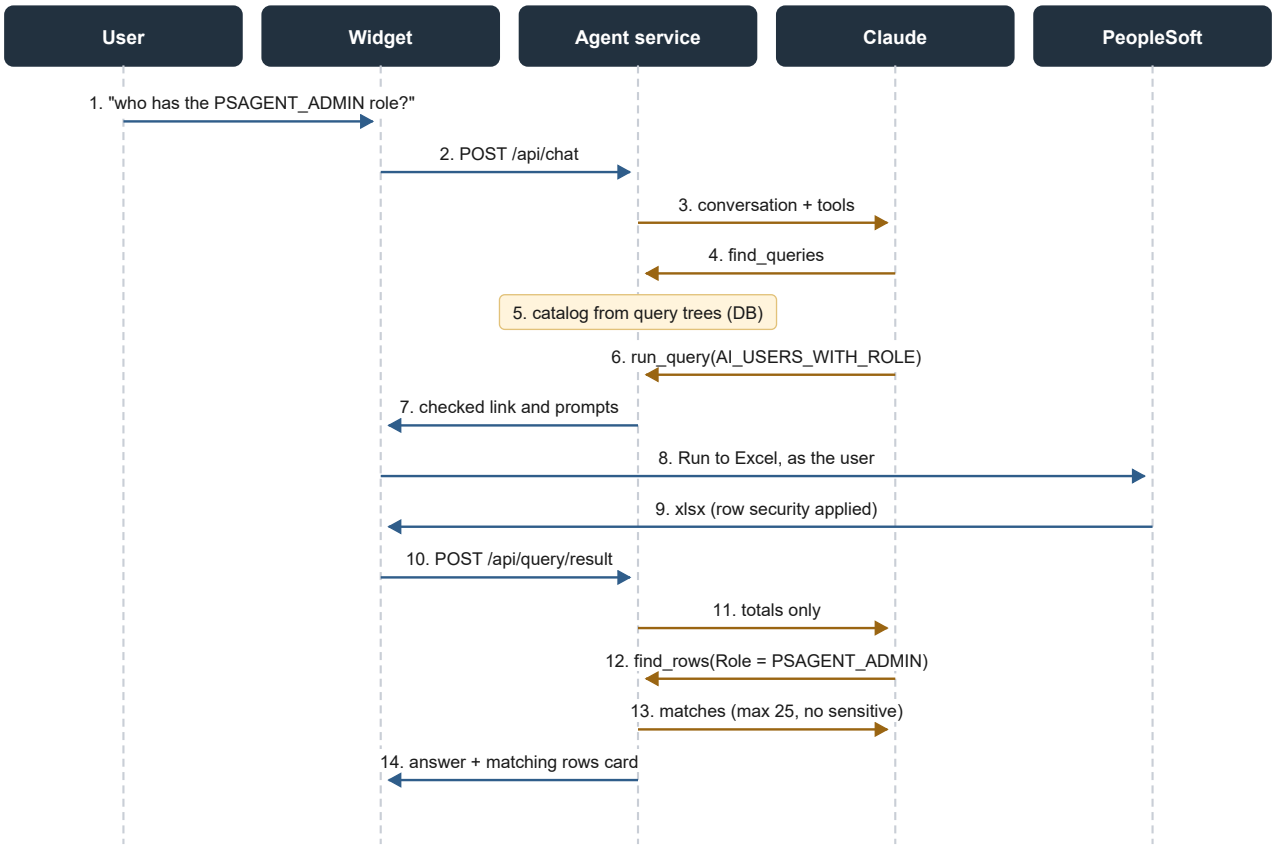


Figure 5-6 Ask the data: Claude picks the query; PeopleSoft runs it as the user

6 Operations

6.1 Installing and deploying

Step	What	Detail
1	Agent service	Copy agent-service/ to the host; pip install -r requirements.txt; create .env (6.2); start with uvicorn.
2	Sign-on iScript	Create WEBLIB_PSAGENT.ISCRIPT1 FieldFormula from psagent_iscript; set &AGENT;_URL and &AGENT;_SECRET (= PSAGENT_SHARED_SECRET); grant the web library to the users' permission lists.
3	Widget	Upload psagent.js through Branding (custom JavaScript object) so it loads on every page; check PSAgent.version in the browser console.
4	Roles	Create PSAGENT_USER, PSAGENT_MONITOR, PSAGENT_ADMIN, PSAGENT_AUDITOR and give them to users.
5	Feature setup	WEBLIB_SOAPTOCI and CI methods for Excel to CI; AI_ queries for Ask the data; PSAGENT_SVC for log downloads; SMTP for email.
6	Verify	Start-up log clean; the live API test passes; run the end-to-end test workbook (6.4).

6.2 Configuration

Setting	Purpose	Default
ANTHROPIC_API_KEY	Claude API key	required
PSAGENT_SHARED_SECRET	Same value as &AGENT;_SECRET in the iScript	required
PSAGENT_ALLOWED_ORIGINS	PeopleSoft address(es) allowed to call the service (CORS)	none (same origin)
PSAGENT_MODEL, PSAGENT_EFFORT	Claude model and effort for chat	claude-opus-5-5, low
PSAGENT_TROUBLESHOOT_EFFORT	Effort for investigations	medium
PSAGENT_DB_DSN, _USER, _PASSWORD, _SCHEMA	Read-only database access (troubleshooting, watcher, CI checks, Ask the data)	off when empty
PSAGENT_CI, PSAGENT_QUERIES, PSAGENT_WATCH, PSAGENT_AUDIT	Feature switches	on
PSAGENT_QUERY_PREFIX	Name prefix of curated queries	AI_
PSAGENT_PS_URL	PeopleSoft address for links and the service user's sign-in	
PSAGENT_SVC_USER, _PASSWORD	Service user for overnight log downloads	
PSAGENT_SMTP_*, PSAGENT_ALERT_TO, PSAGENT_AUDIT_TO	Email: server and recipients	no email
PSAGENT_WATCH_MAX_INVESTIGATIONS, _SUMMARY	Daily investigation cap; daily summary time	20, 07:00
PSAGENT_AUDIT_DAYS, PSAGENT_BUDGET, PSAGENT_AUDIT_WEEKLY	Audit retention; monthly budget in USD; weekly email	90, none, Mon 08:00
PSAGENT_TEST_ENV, PSAGENT_TEST_USER, _PASSWORD	Regression tests (test systems only); unattended runs	off

Setting	Purpose	Default
PSAGENT_DEV_OPRID, PSAGENT_DEV_ROLES	Development only: run without a token. Refused with the secret.	unset

6.3 Cost

Measured on Claude Opus 5.5 at list prices. Recorded tasks, the watcher's routine checks, the audit dashboard and Excel to CI loads use little or no Claude.

Action	Claude cost	Time
Recorded task run	about $0.02	seconds
Live mode, per page step	about $0.015 to $0.02	
Process investigation	a few tens of cents	20 to 60 s
Security access check / absence review	similar to an investigation	25 to 45 s
Runbook draft / watch plan from a template	a few cents to tens of cents	about 30 s
Watcher, routine check	none (one read-only query)	
Watcher, new failure	like an investigation	about 45 s
Ask the data	two to three chat turns	
Excel to CI load	none	about 1 s per row
Admin dashboard	none (local SQLite)	

6.4 Testing and quality

Level	What it covers	Count
Unit tests	Every module, each role through the API, refused attempts, masking, prompts, CI checks, 457 real CI shapes	232
Live API test	The real Claude API accepts every tool list (strict-tool and grammar limits); run on request	1 (in the 232)
Browser tests	The widget end to end against mock PeopleSoft pages, a mock SOAPTOCI and a scripted Claude	23
End-to-end test workbook	Manual tests on the real system with real Claude, by role (PSAgent_E2E_Test_Cases.xlsx)	110 cases (52 P1)

Why the manual end-to-end tests still matter
Automated tests use a scripted Claude and mock pages, so they cannot see limits of the real API, real PeopleSoft behaviour or real data. One example: the real API refused a request with too many strict tools, which no unit test could detect; the live API test now covers it.

6.5 Running the service

- **Start-up log:** lists the allowed origins, the roles model, the database, the features that are on, the loaded runbooks and warnings for unused or development-only settings.
- **Audit dashboard:** usage, cost and failures; refused attempts show misconfigured roles or probing.
- **Scaling:** one worker with in-memory sessions today; shared sessions (Redis) and several workers are planned.

7 Limitations and roadmap

7.1 Open items before production

Item	Why it matters
Dedicated read-only database account (PSAGENT_RO)	The service uses SYSADM today; read-only is enforced by the service, but the account should not be able to write at all.
Shorter token lifetime	Tokens last a day; one hour with silent refresh makes role changes and leaked tokens expire quickly.
PSAGENT_SVC permissions and password	Remove PSADMIN; keep Process Monitor and Report Manager only; change the password used in testing.
Query Viewer link and query trees	The run link did not check query trees for a PSADMIN user; retest with a normal user.
SMTP server	Needed for alert, summary, budget and weekly emails.
PSAGENT_AUDITOR role and the updated iScript	Needed for the admin dashboard.
Hosting	Replace ngrok with a reverse proxy under the PeopleSoft host; keep keys in a vault.
Real Excel to CI Create and Update	Proven read-only so far; run on a test system first.

7.2 Known limitations

- Classic pages only for tasks, live mode and the recorder (no Fluid yet).
- The troubleshooter is strongest on Application Engine; SQR and COBOL source and Integration Broker failures are not covered.
- The security troubleshooter does not check query, process or Finance row security.
- Runbook approvals are clicks in the same browser, not routed to the approver; a run does not survive closing the browser.
- Chat history lasts only for the browser session; one service worker.

7.3 Roadmap

Area	Next
Fix and rerun	After an alert and its diagnosis, propose a known fix; Rerun after a person's click; the watcher confirms and closes the incident.
Excel to CI	Large files through the Process Scheduler with a process instance; bulk loads for recorded tasks; matching columns of any spreadsheet.
Ask the data	A query result as a watchdog ("tell me when this changes"); favourite questions on the welcome screen.
Security	Apply for row-security refresh; compare two users' access; access reviews and an access request workflow.
Runbooks	Edit in the chat; real approvals by email or Teams; resume and hand-off; version history; evidence packs for auditors.
Watcher	Integration Broker and long-running processes; pay cycle deadlines; Teams; learning what is normal.

Area	Next
Platform	Fluid pages; shared sessions and several workers; production hosting and a security review.

Appendix A Glossary

Term	Meaning
AE	Application Engine, PeopleSoft's batch program framework
Branding	PeopleTools feature used to load custom JavaScript on every page
CI	Component Interface: PeopleSoft's API to a component, used by ExcelToCI and SOAPTOCI
Confirm card	The widget's summary of a task's values; nothing runs until the user clicks Confirm
iScript	A PeopleCode function callable by URL in the user's session (WEBLIB_ records)
Live mode	Claude operating PeopleSoft one page step at a time
Query trees	PeopleSoft query security: permission lists to access groups to records
Runbook	A published checklist of checks, tasks, processes, approvals and manual steps
SJT tables	HCM row-security tables (for example SJT_OPR_CLS) that decide which people a user sees
SOAPTOCI	The delivered WEBLIB_SOAPTOCI iScript that ExcelToCI uses to call Component Interfaces
Strict tool	A tool whose input shape the Claude API guarantees to match its schema
Token	The signed proof of the user's identity and agent roles, issued by WEBLIB_PSAGENT
Watchdog / watch plan	A rule the watcher checks on its own / a saved set of watchdogs

Appendix B HTTP API summary

Every call needs `Authorization: Bearer <token>` and is recorded in the audit log under the user and the feature its path belongs to.

Area	Endpoints
Chat and tasks	POST /api/chat, /api/confirm, /api/result, /api/batch/result; GET /api/me
Live mode	POST /api/live/step, /api/live/recover
Investigations	POST /api/troubleshoot/step, /api/approvals/review, /api/security/apply
Recorder (ADMIN)	POST /api/recordings/draft, /api/recordings/save
Runbooks	POST /api/runbooks/save (ADMIN), /api/runbooks/step
Watcher (MONITOR)	POST /api/monitors/save, /api/watchplans/save, /api/watchdogs/action; GET /api/watchdogs, /api/incidents, /api/watchplans/export
Excel to CI	GET /api/ci, /api/ci/shape, /api/ci/loads; POST /api/ci/template, /api/ci/check, /api/ci/start, /api/ci/next, /api/ci/result
Ask the data	GET /api/queries; POST /api/query/result
Audit (AUDITOR)	GET /api/audit, /api/audit/export
Regression (ADMIN, test systems)	GET /api/regress; POST /api/regress/save, /start, /next, /result, /live, /stop, /unattended
Health	GET /health